Homework 3

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From previous study we have

$$\frac{\partial^2 \phi}{\partial t^2} - c^2 \nabla^2 \phi = F \tag{1}$$

Set $\mathcal{L} = \alpha(x)\dfrac{\partial^2}{\partial t^2} - c^2\nabla^2$ and $\mathcal{L}G(\mathbf{x}, t; \mathbf{y}; \tau) = \delta(t-\tau)\delta^n(\mathbf{x}-\mathbf{y})$, in which $\alpha = 1$, we have

$$\left(\frac{\partial^2}{\partial t^2} - c^2\nabla^2\right) G(\mathbf{x}, t; \mathbf{y}; \tau) = \delta(t-\tau)\delta^n(\mathbf{x}-\mathbf{y}) \tag{2}$$

By using initial conditions

$$\phi(x, 0) = 0, \tag{3}$$

$$\frac{\partial \phi}{\partial t}(x, 0) = 0. \tag{4}$$

Different equations of Eq .(2) 's type can be solved by the general solution (Skinner . Eq (7.3))

$$\phi(\mathbf{x}, t) = \int_0^\infty \int_{\mathbb{R}^n} F(\mathbf{y}, \tau) G(\mathbf{x}, t; \mathbf{y}; \tau) d^n y \, d\tau, \tag{5}$$

the Fourier transform of Eq. (5) with respect to the spatial variables $\mathbf{x}$ can be obtained by

$$\mathcal{F}\left[\frac{\partial^2}{\partial t^2}G(\mathbf{x},t;\mathbf{y};\tau)\right] = \frac{\partial^2}{\partial t^2}\tilde{G}(\mathbf{x},t;\mathbf{y};\tau), \tag{6}$$

$$-c^2\mathcal{F}\left[\nabla^2 G(\mathbf{x},t;\mathbf{y};\tau)\right] = -c^2\int_{-\infty}^{+\infty} e^{-ikx}\nabla^2 G(\mathbf{x},t;\mathbf{y};\tau) = k^2c^2\tilde{G}(\mathbf{x},t;\mathbf{y};\tau), \tag{7}$$

$$\mathcal{F}\left[\delta^n(\mathbf{x}-\mathbf{y})\right] = \int \delta^n(\mathbf{x}-\delta(t-\tau)\mathbf{y})e^{-i\mathbf{k}\cdot\mathbf{x}}d^nx = \delta(t-\tau)e^{-i\mathbf{k}\cdot\mathbf{y}}(\delta \text{ Dirac props.}) \tag{8}$$

We may now write Eq .(2) as

$$\frac{\partial^2}{\partial t^2}\tilde{G}(\mathbf{x},t;\mathbf{y};\tau) + k^2c^2\tilde{G}(\mathbf{x},t;\mathbf{y};\tau) = e^{-i\mathbf{k}\cdot\mathbf{y}}\delta(t-\tau). \tag{9}$$

This equation have been mentioned in Eq .(7.51) in Skinner

$$\mathcal{L}y = f(t). \tag{10}$$

A general solution for this type equation for $t > \tau$ is

$$y = A\cos\Big(\sqrt{k^2c^2}t\Big) + B\sin\Big(\sqrt{k^2c^2}t\Big) \tag{11}$$

By applying initial conditions at $t = \tau$. Since $G = 0$ at $t < \tau$, we obtain

$$G(t,\tau) = A\cos\Big(\sqrt{k^2c^2}t\Big) + B\sin\Big(\sqrt{k^2c^2}t\Big) = 0 \to C = 0 \tag{12}$$

$$G^{'}(t,\tau) = \frac{1}{\alpha} \to \sqrt{k^2c^2}B\cos 0 = 1 \Rightarrow B = \frac{1}{\sqrt{k^2c^2}} \tag{13}$$

in which $\alpha = 1$. We now write Green function as

$$\tilde{G}(\mathbf{k},t;\mathbf{y},\tau) = \Theta(t-\tau)e^{-i\mathbf{k}\cdot\mathbf{y}}\frac{\sin\big(kc(t-\tau)\big)}{ck}. \tag{14}$$

Notice that we have parameterized k then treat t as our variable in the initial value problem, for symplicity we write $k \equiv |\mathbf{k}|$ which will be explain in next compute.

We now may write Eq .(14) in the inverse Fourier transform form

$$\begin{aligned} G(\mathbf{x},t;\mathbf{y};\tau) &= \frac{1}{(2\pi)^n}\int_{\mathbb{R}^n} e^{i\mathbf{k}\cdot\mathbf{x}}\tilde{G}(\mathbf{k},t;\mathbf{y},\tau)d^nk \\ &= \frac{1}{(2\pi)^n}\int_{\mathbb{R}^n} e^{i\mathbf{k}\cdot\mathbf{x}}\Theta(t-\tau)e^{-i\mathbf{k}\cdot\mathbf{y}}\frac{\sin\big(kc(t-\tau)\big)}{ck}d^nk \\ &= \frac{\Theta(t-\tau)}{(2\pi)^n}\int_{\mathbb{R}^n} e^{i\mathbf{k}\cdot(\mathbf{x}-\mathbf{y})}\frac{\sin\big(kc(t-\tau)\big)}{ck}d^nk. \end{aligned} \tag{15}$$

For symplicity, we choose $n = 3$, so that

$$|\mathbf{k}| = k, \tag{16}$$

$$\mathbf{k} \cdot (\mathbf{x} - \mathbf{y}) = k|\mathbf{x} - \mathbf{y}| \cos\theta. \tag{17}$$

Thus, we may write the Green function in spherical symmetry, Eq .(15) becomes

$$\begin{aligned}
G &= \frac{\Theta(t-\tau)}{(2\pi)^3} \int_0^\infty \int_0^\pi \int_0^{2\pi} e^{ik|\mathbf{x}-\mathbf{y}|\cos\theta} \frac{\sin kc(t-\tau)}{ck} k^2 d\phi d\theta dk \\
&= \frac{\Theta(t-\tau)}{c(2\pi)^2} \int_0^\infty \int_0^\pi e^{ik|\mathbf{x}-\mathbf{y}|\cos\theta} \sin kc(t-\tau) k d\theta dk \\
&= \frac{\Theta(t-\tau)}{c(2\pi)^2} \int_0^\infty \left[\int_{-1}^1 e^{ik|\mathbf{x}-\mathbf{y}|\cos\theta} d\cos\theta \right] \sin kc(t-\tau) k dk \\
&= \frac{\Theta(t-\tau)}{c(2\pi)^2} 2\pi i c|\mathbf{x}-\mathbf{y}| \int e^{ik|\mathbf{x}-\mathbf{y}|} \sin kc(t-\tau) k dk
\end{aligned} \tag{18}$$

The intergral term has the form of inverse Fourier transform, expand the sin function as

$$\sin\big(kc(t-\tau)\big) = \frac{1}{2i} \left(e^{ikc(t-\tau)} - e^{-ikc(t-\tau)} \right), \tag{19}$$

plugging into the intergral term we have

$$\begin{aligned}
\int e^{ik|\mathbf{x}-\mathbf{y}|} \sin kc(t-\tau) k dk &= \frac{1}{2i} \int e^{ik|\mathbf{x}-\mathbf{y}|+c(t-\tau)} k dk - \frac{1}{2i} \int e^{ik|\mathbf{x}-\mathbf{y}|-c(t-\tau)} k dk \\
&= -2\pi i \delta(|\mathbf{x}-\mathbf{y}| + c(t-\tau)) + 2\pi i \delta(|\mathbf{x}-\mathbf{y}| - c(t-\tau)),
\end{aligned} \tag{20}$$

therefore the Green fucntion now is

$$G = \frac{\Theta(t-\tau)}{4\pi c|\mathbf{x}-\mathbf{y}|} \big[\delta(|\mathbf{x}-\mathbf{y}| + c(t-\tau)) - \delta(|\mathbf{x}-\mathbf{y}| - c(t-\tau)) \big]. \tag{21}$$

notice that in the delta Dirac function the argument already ensures that $t - \tau > 0$ so we do not neither the Θ-function nor the first δ-function

$$G = \frac{1}{4\pi c|\mathbf{x}-\mathbf{y}|} \big[-\delta(|\mathbf{x}-\mathbf{y}| - c(t-\tau)) \big] \tag{22}$$

Therefore, the general solution to $\mathcal{L}\phi = F$ obeying initial conditions is

$$\begin{aligned}
\phi(\mathbf{x}, t) &= \int_0^t \frac{1}{4\pi c} \int_{\mathbb{R}^n} G F(\mathbf{y}, \tau) d^n y d\tau \\
&= -\frac{1}{4\pi c^2} \int_{\mathbb{R}^n} \frac{F(\mathbf{y}, \tau)}{|\mathbf{x}-\mathbf{y}|} \delta\big(|\mathbf{x}-\mathbf{y}| - c(t-\tau) \big) d^n y \, d\tau
\end{aligned} \tag{23}$$

This is the solution of forced wave equation in 3+1 dimensions. The general solution to the forced wave equation with homogeneous boundary conditions in 1+1 dimensions can be obtained by re-using Eq .(15) with choosing $n = 1$

$$G(x,t;y,\tau) = \frac{\Theta(t-\tau)}{2\pi} \int e^{ik(x-y)} \frac{\sin\big(kc(t-\tau)\big)}{ck} dk, \tag{24}$$

expand the sin function as

$$\sin\big(kc(t-\tau)\big) = \frac{1}{2i}\left(e^{ikc(t-\tau)} - e^{-ikc(t-\tau)}\right) \tag{25}$$

we now have

$$\begin{aligned} G(x,t;y,\tau) &= \frac{\Theta(t-\tau)}{4ic\pi} \int \frac{1}{k}\left(e^{ik(x-y+c(t-\tau))} + e^{ik(x-y+c(t-\tau))}\right) dk \\ &= \frac{\Theta(t-\tau)}{4ic\pi} 2\pi i \left[\delta(|x-y| + c(t-\tau)) - \delta(|x-y| - c(t-\tau))\right] \\ &= \frac{1}{2c}\left[\delta(|x-y| + c(t-\tau)) - \delta(|x-y| - c(t-\tau))\right] \end{aligned} \tag{26}$$